# Uniform decay of every Fourier mode of the density under free Schrödinger evolution

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*Editorial note.* The counted statement is the author's labelled proposition 'propfreephasemixing' (source lines 550-555); the proof environment begins on the next line after \end{proposition} apart from one blank line, so the pairing is by adjacency.

*Editorial note.* Five context spans are retained, carrying the author's toolbox of quantum operators, the commutation remark, the two Wigner-transform identities and the free Schrödinger equation; every label used inside the retained text is therefore defined inside the package.

*Editorial note.* The retained preamble is the publisher layout adapter plus the native macro, operator and theorem block of the same pinned source.

*Editorial note.* No retained line refers to a label, citation or figure outside the retained spans, so the package declares no external reference.

*Editorial scope.* One original result of the pinned publisher article, counted once: uniform decay of the Fourier modes of the density under free Schrödinger evolution (author label propfreephasemixing).

*Allowed standard prerequisites (editorial).*

- Free Schrödinger evolution. Q_FS(t) = e^{i hbar t Delta / 2} Q_in e^{-i hbar t Delta / 2} is the solution of the free Schrödinger equation.. It defines the density whose Fourier modes are estimated, and its equation is retained as context.
- Commutation of the quantum operators. The quantum derivatives and position operators commute in the four combinations recorded in the author's remark.. It allows the identities for the individual operators to be extended to the joint weight at source line 557.
- Weighted Sobolev norm H^{N0}_{M,M}. The norm controls the kernel up to order N0 in both variables with the weight M = ceil((d+1)/2).. It is the norm on the right-hand side of the estimate.

**Definition 0.1** (Quantum derivatives and weights)**.** *The quantum analogues of spatial and velocity derivatives are*

$$\boldsymbol{\nabla}_x Q := [\nabla, Q], \quad (\boldsymbol{\nabla}_x Q)(x,y) = (\nabla_x + \nabla_y)Q(x,y),$$
$$\boldsymbol{\nabla}_\xi Q := [\tfrac{x}{i\hbar}, Q], \quad (\boldsymbol{\nabla}_\xi Q)(x,y) = (\tfrac{x-y}{i\hbar})Q(x,y).$$

*The quantum analogues of spatial and velocity weights are*

$$\begin{aligned}\boldsymbol{x}Q &:= \tfrac{1}{2}(xQ + Qx), \qquad (\boldsymbol{x}Q)(x,y) = (\tfrac{x+y}{2})Q(x,y),\\ \boldsymbol{\xi}Q &:= -\tfrac{i\hbar}{2}(\nabla Q + Q\nabla), \quad (\boldsymbol{\xi}Q)(x,y) = -\tfrac{i\hbar}{2}(\nabla_x - \nabla_y)Q(x,y).\end{aligned}$$

*Remark* 0.2 (Commutation of the quantum operators). The operators of Definition 0.1 satisfy the commutation relations

$$[\boldsymbol{\nabla}_x, \boldsymbol{\nabla}_\xi] = [\boldsymbol{x}, \boldsymbol{\xi}] = [\boldsymbol{\nabla}_x, \boldsymbol{\xi}] = [\boldsymbol{x}, \boldsymbol{\nabla}_\xi] = 0,$$

in exact analogy with the classical operators $\nabla_x$, $\nabla_\xi$, $x$, $\xi$ acting on functions $f(x,\xi)$. Indeed, on the level of integral kernels, $\boldsymbol{x}$ and $\boldsymbol{\nabla}_x$ involve only the sum $x+y$, while $\boldsymbol{\xi}$ and $\boldsymbol{\nabla}_\xi$ involve only the difference $x-y$: for instance,

$$(\nabla_x - \nabla_y)\left[\left(\tfrac{x+y}{2}\right)Q(x,y)\right] = \left(\tfrac{x+y}{2}\right)(\nabla_x - \nabla_y)Q(x,y),$$

which gives $[\boldsymbol{x}, \boldsymbol{\xi}] = 0$, and similarly $(\nabla_x + \nabla_y)(x-y) = 0$ gives $[\boldsymbol{\nabla}_x, \boldsymbol{\nabla}_\xi] = 0$. The remaining relations hold since $\boldsymbol{\nabla}_x$ and $\boldsymbol{\xi}$ are both constant-coefficient differential operators, and $\boldsymbol{x}$ and $\boldsymbol{\nabla}_\xi$ are both multiplication operators. In particular, products such as $\boldsymbol{\xi}^\alpha \boldsymbol{x}^\beta$ are independent of the order of the factors. By contrast, $[\boldsymbol{\nabla}_x, \boldsymbol{x}] \neq 0$ and $[\boldsymbol{\nabla}_\xi, \boldsymbol{\xi}] \neq 0$, again as in the classical case.

**Lemma 0.3.** *Suppose $Q \in \mathcal{H}^0_{M,M}(\mathbb{R}^d;\mathbb{C})$. Then the Wigner transform of $Q$ satisfies, for all $k, \eta \in \mathbb{R}^d$,*

$$\left|\widehat{W[Q]}(k,\eta)\right| \lesssim_d \|Q\|_{\mathcal{H}^0_{M,M}}.$$

**Lemma 0.4.** *Suppose $Q \in \mathcal{L}^2(\mathbb{R}^d;\mathbb{C})$. Then the Wigner transform of $Q$ satisfies the following:*

$$\begin{aligned}
\nabla_x W[Q](x,\xi) &= W[\boldsymbol{\nabla}_x Q](x,\xi), & ik\widehat{W[Q]}(k,\eta) &= \widehat{W[\boldsymbol{\nabla}_x Q]}(k,\eta),\\
\nabla_\xi W[Q](x,\xi) &= W[\boldsymbol{\nabla}_\xi Q](x,\xi), & i\eta\widehat{W[Q]}(k,\eta) &= \widehat{W[\boldsymbol{\nabla}_\xi Q]}(k,\eta),\\
xW[Q](x,\xi) &= W[\boldsymbol{x}Q](x,\xi), & i\nabla_k\widehat{W[Q]}(k,\eta) &= \widehat{W[\boldsymbol{x}Q]}(k,\eta),\\
\xi W[Q](x,\xi) &= W[\boldsymbol{\xi}Q](x,\xi), & i\nabla_\eta\widehat{W[Q]}(k,\eta) &= \widehat{W[\boldsymbol{\xi}Q]}(k,\eta).
\end{aligned}$$

$$\begin{cases} i\hbar\partial_t Q_{\mathrm{FS}} = \left[-\dfrac{\hbar^2}{2}\Delta, Q_{\mathrm{FS}}\right], \\ Q_{\mathrm{FS}}(0) = Q_{\mathrm{in}}, \end{cases} \tag{1}$$

**Proposition 0.5** (Phase mixing for the free Schrödinger equation)**.** *Fix $N_0 \geq 0$. Then for any $t \geq 0$ and $k \in \mathbb{R}^d$, the density of the solution $Q_{\mathrm{FS}}$ to* (1) *satisfies*

$$\left|\widehat{\rho_{Q_{\mathrm{FS}}}}(t,k)\right| \lesssim_d \frac{\|Q_{\mathrm{in}}\|_{\mathcal{H}^{N_0}_{M,M}}}{\langle k, kt\rangle^{N_0}}.$$

*Proof.* First recall that $\widehat{\rho_{Q_{\mathrm{FS}}}}(t,k) = \widehat{W[Q_{\mathrm{in}}]}(k,kt)$. Since $\boldsymbol{\nabla}_x$ and $\boldsymbol{\nabla}_\xi$ commute (Remark 0.2), the relations of Lemma 0.4 extend from the individual operators to the joint weight $\langle \boldsymbol{\nabla}_x, \boldsymbol{\nabla}_\xi\rangle^{N_0}$, whence

$$\langle k, kt\rangle^{N_0}\widehat{\rho_{Q_{\mathrm{FS}}}}(t,k) = \langle k,\eta\rangle^{N_0}\widehat{W[Q_{\mathrm{in}}]}(k,\eta)\Big|_{\eta=kt} = \widehat{W\left[\langle \boldsymbol{\nabla}_x, \boldsymbol{\nabla}_\xi\rangle^{N_0} Q_{\mathrm{in}}\right]}(k,kt).$$

The result then follows from Lemma 0.3. □

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